Bishal Dhungana

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SUMMARY

Geomatics Engineer with hands-on experience in UAV mapping (FAA Part 107 certified), GNSS/GPS survey, and LiDAR point cloud processing. I currently lead QA/QC of LiDAR-derived data for electric utility infrastructure, verifying positional accuracy and processing point clouds to support pole hardening and grid resilience work. My background spans field survey, terrain modeling, and geodatabase design, from control surveys and UAV topographic mapping in Nepal to production LiDAR workflows for a major U.S. utility.

SKILLS

Field: GNSS / GPS Survey, UAV Survey (FAA Part 107), Differential GNSS, Total Station and Leveling Survey, Geodatabase Design

Geospatial: ArcGIS Pro / Enterprise / Online, Utility Network, PostGIS, DuckDB Spatial, GDAL, PDAL, GeoPandas, Shapely, PyProj, LiDAR / Point Cloud, Global Mapper, LAStools, Overture Maps, Cloud-Optimized GeoTIFF, Folium, FME

Data Engineering: Python, SQL, ETL/ELT Pipelines, PySpark, DuckDB, Databricks, Data Lakes, Parquet

Cloud & DevOps: AWS Lambda, S3, SageMaker, ECR / ECS, AWS CDK, Docker, IAM, VPC / Network Security, CI/CD

EXPERIENCE

Data Scientist (Geospatial & AI) — Apr 2025 – Aug 2026

Florida Power & Light - Contract, West Palm Beach, FL

- Built a storm-pilot deployment application in ArcGIS Experience Builder to manage internal and external UAV pilots, with real-time location tracking, multi-deployment coordination, time-of-deployment logging, and multi-level access for enterprise users.
- Built an API-based metadata pipeline with spatial (10m/30ft) and temporal (30-day) joins matching 350K+ condition reports to an 8M-image catalog, enriching ~2M images with condition metadata.
- Built and deployed GIS applications (ArcGIS Enterprise, Experience Builder, Dashboards, Utility Network) for infrastructure monitoring and disaster response, including spatial web portals used in real time during Hurricane Milton for structural impacts, outages, and crew deployment.

Associate Geospatial Analyst — Sep 2024 – Mar 2025

Florida Power & Light - Contract, Jupiter, FL

- Led QA/QC of LiDAR-derived power grid data for infrastructure resilience work, including pole hardening.
- Managed PostgreSQL/PostGIS databases, optimized Cloud-Optimized GeoTIFFs, and automated QA workflows in FME, ArcPy, and Model Builder to streamline data storage and validation.
- Built a suite of LiDAR-based transmission corridor tools: clearance and right-of-way mapping (GeoPandas, pykml, ArcPy); a point-cloud classification tool extracting pole height, length, top/bottom diameter, tilt, wire span, and communication attachment points, used on a Key West fiber attachment project and exported as visualized KML; a CLI for vegetation-density classification (low/medium/high) supporting vegetation management at FPL's Turkey Point nuclear site; and a terrain-model generation CLI (GDAL, PDAL, GeoPandas) built against corridor LiDAR.

Research Assistant — Jan 2023 – Aug 2024

Florida Atlantic University, Boca Raton, FL

- Conducted FEMA hydrologic and hydraulic (H&H) flood risk assessments for 3 Florida communities using Python automation, part of FAU's $1.7M FEMA-funded CWR3 Watershed Master Planning Initiative to build a statewide planning template.
- Built an automated ArcGIS Pro geoprocessing pipeline, cutting processing time by 50%.
- Built a Python geocoding tool processing 3 million addresses, replacing a paid service; cut turnaround from 3 months to 1 week and saved $20K in licensing.

Geomatics Engineer — Dec 2020 – Dec 2022

Freelance Consultant, Kathmandu, Nepal

- Led survey teams for municipal road, control, and construction surveys, including GNSS landmark and UAV topographic/transmission-line surveys, and drone photogrammetry for orthomosaic mapping.
- Designed municipal geodatabases (land use, soil, cadastral, resource maps) with topology and attribute rules.
- Served as GIS expert for Nepal's Survey Department land-use hand-over program, training 40+ officials in spatial analysis and database management.

EDUCATION

MSc, Geosciences — GIS Concentration — Florida Atlantic University · 2023–2024

BSc, Geomatics Engineering — Tribhuvan University, Nepal · 2016–2021

CERTIFICATIONS & TRAINING

- The Rise of the AI Data Engineer Bootcamp (2026)
- FAA Part 107 — Remote Pilot Certificate
- Machine Learning Specialization — DeepLearning.AI
- Deep Learning Specialization — DeepLearning.AI
- Remote Sensing Image Acquisition & Analysis — UNSW Canberra
- GIS Mapping & Spatial Analysis — U. Toronto
- Data Science with R — Johns Hopkins
- GIS Specialization — UC Davis
- Introduction to Databases — Meta
- **NASA ARSET Certifications (12):** Applied remote sensing training (2022-2024) in climate, disaster, and environmental risk modeling.
- **Esri Training Certifications (8):** GIS fundamentals, cartography, spatial data science, and imagery analysis.

PUBLICATIONS

Dhungana, B.; Liu, W. *Urban–Rural Exposure to Flood Hazard and Social Vulnerability in the Conterminous United States*. ISPRS Int. J. Geo-Inf. 2024, 13, 339.

Mandal, A.K.; Thapa, A.K.; Thapa Chhetri, M.; Dhungana, B.; Bloetscher, F.; Yong, Y.; Su, H. *PyWMP: A Unified Python Framework for Multi-Fidelity 1D, 2D, and Hybrid Event-Based Flood Modeling*. Manuscript in review, DOI pending; ongoing contributor to the open-source PyWMP package.

PROJECTS

3DEP LiDAR Terrain Builder: Generates DEM, DSM, hillshade, slope, aspect, and contour products from USGS 3DEP lidar. Draw an area of interest in the browser; it generates a configured local script — no data ever uploads. *(PDAL, GDAL, RichDEM, AWS EPT/S3)* Demo | Repo

3d-export: 3D-style land cover export from Sentinel-2 tiles, clipped to any country and automated via GitHub Actions — a modernized take on the rayshader R workflow. Demo | Repo

Road-Centerline: A published Python package that extracts road centerlines from polygon geometries — worldwide, in any input CRS, in any format GeoPandas can read. CLI and Python API, computed via medial-axis skeletonization. *(Python, GeoPandas, pygeoops, PyProj, PyPI)* Repo

fflood-nep: A reproducible flood-response pipeline for Nepal's August 2026 Rasuwa/Bhote Koshi flash flood — runs Sentinel-1 SAR change detection for flood extent, scores building/road/facility exposure by municipality, and tracks a separate InSAR pipeline for landslide/avalanche source detection. Built and shipped inside the event's first week. *(Python, Sentinel-1 SAR, STAC, GDAL, InSAR)* Demo | Repo